

Software Requirements Specification (SRS)

Name: CrowdSenseNet

Design and Implementation of a Crowdsensed Drive Test Mobile Application for Coverage Prediction

5.1 Introduction

5.1.1 Purpose

This document provides the complete Software Requirements Specification (SRS) . The purpose of this document is to specify the requirements for **CrowdSenseNet**, a system designed to address the lack of transparent network coverage data in Cameroon. By leveraging crowdsensing technology, the system aims to empower both consumers and network operators with real-time, accurate signal maps, replacing expensive traditional drive-tests with a community-driven mobile solution.”

5.1.2 Document Conventions

- **Must Have:** Critical requirements for the Minimum Viable Product (MVP)
- **Should Have:** Important features that add significant value but are not blocking
- Signal measurements are expressed in **dBm** for power and **dB** for quality

5.1.3 Intended Audience

- **Developers and architects:** focus on Sections 5.3 (System Features) and 5.7 (Architecture)
- **Network engineers:** focus on Section 5.1.5 (Definitions) and Section 5.6 (Data Requirements)
- **Academic supervisors:** focus on Section 6 (Stakeholder Validation) and the traceability matrix at the end

5.1.4 Project Scope

CrowdSenseNet is an Android application that turns ordinary smartphones into **Class-2 IoT sensing devices**. It passively collects cellular metrics (RSRP, RSRQ, SINR) to build real-time coverage heatmaps and predict coverage holes (RSRP below -110 dBm) using a machine learning model. **iOS support is excluded** in this phase due to API restrictions and scope limitations.

The Minimum Viable Product (MVP) covers: signal data collection (RSRP, RSRQ, SINR, GPS), local offline storage, backend data upload, a basic coverage map, and ML-based coverage hole prediction.

5.1.5 Key Definitions

TERM	DEFINITION
RSRP	Reference Signal Received Power. The primary LTE signal strength metric. Coverage hole threshold: below -110 dBm
RSRQ	Reference Signal Received Quality. Measures signal quality relative to interference
SINR	Signal to Interference-plus-Noise Ratio. Indicates signal clarity
COVERAGE HOLE	Any location where predicted RSRP is below -110 dBm
UUID	Universally Unique Identifier. A random anonymous token generated on first launch
CROWDSENSING	Collecting data passively from many users to build a picture of real-world conditions
DRIVE TEST	Traditional method of measuring mobile coverage using engineers with specialised equipment in a vehicle

Table 5.1: Key definitions and terminology

5.2 Overall Description

5.2.1 Product Perspective

CrowdSenseNet is a standalone Android application operating within a three-tier architecture:

- **Tier 1 (Client):** Android app written in Kotlin for data collection and local SQLite storage
- **Tier 2 (Backend):** Python FastAPI server for data validation and PostgreSQL storage
- **Tier 3 (Analytics):** Python ML model for coverage prediction and heatmap generation

5.2.2 User Classes

- **General Contributors:** Non-technical users who passively contribute signal data.
- **Network Engineers:** High-technical users at MTN/Orange who access maps to fix problem areas.
- **System Administrators:** Experts managing the backend, database, and ML updates.

5.2.3 Operating Environment

- **Mobile platform:** Android 8.0 (API level 26) minimum
- **Local database:** Room (SQLite-based)
- **Remote database:** PostgreSQL with PostGIS for spatial data
- **Network:** 4G, 5G, or WiFi for data synchronisation

5.2.4 Constraints

- **Battery constraint:** Adaptive sampling is required to prevent unacceptable drain
- **Geographic scope:** Limited to the Cameroon bounding box (Latitude 1.65N to 13.08N)
- **Permission dependency:** Requires ACCESS_FINE_LOCATION and READ_PHONE_STATE permissions

5.2.5 Assumptions and Dependencies

- Users must grant required hardware permissions on first launch
- A minimum of 50 readings per 1 km squared is required before the ML model can generate a prediction
- Android TelephonyManager API must support SINR on the user's device (varies by Android version and chipset)

5.3 System Features

5.3.1 Functional Requirements

ID	Requirement	Priority
FR-01	Single-tap session start/stop from the Home screen	Must Have
FR-02	Auto-pause collection when the device is stationary for more than 30 seconds	Must Have
FR-03	Auto-resume collection when device speed exceeds 5 km/h	Must Have
FR-04	Capture RSRP, RSRQ, SINR, and RSSI via TelephonyManager API	Must Have
FR-05	Capture GPS coordinates with every signal reading	Must Have
FR-06	Discard readings where GPS accuracy is worse than 15 metres	Must Have
FR-07	Store all readings immediately in local Room database	Must Have
FR-08	Generate and store a random anonymous UUID on first launch	Must Have
FR-09	Upload data to server over WiFi (default)	Should Have
FR-10	Allow user to enable mobile data upload via Settings toggle	Should Have
FR-11	Retry failed uploads automatically every 30 minutes	Should Have
FR-12	Export collected data in GeoJSON and CSV formats	Should Have
FR-13	Render a coverage heatmap on the Map screen	Should Have

FR-14	Display a real-time signal dashboard on the Home screen	Could Have
FR-15	Allow user to adjust sampling interval (5 to 30 seconds) in Settings	Could Have
FR-16	Tag readings as indoor or outdoor based on sensor data	Could Have

Table 5.3: Functional requirements with IDs and priorities

5.4 External Interface Requirements

5.4.1 User Interface

- **Modern Design:** Built with Material Design 3 and responsive ConstraintLayouts
- **Four core screens:** Home (Dashboard), Map (Heatmap), History, and Settings
- All key actions must be reachable within 3 taps
- Supports English and French based on device language settings

5.4.2 Hardware Interfaces

- **GPS:** FusedLocationProvider API for battery-efficient location tracking
- **Cellular modem:** TelephonyManager for hardware-level signal parameter access

5.4.3 Software Interfaces

- **Maps:** Google Maps API or OpenStreetMap for the heatmap layer
- **Server API:** REST over HTTPS for all app-to-server communication
- **Local DB:** Room (Android) linked to SQLite for offline storage

5.4.4 Communication Interfaces

- All transmissions must use TLS 1.2 or higher
- WiFi-only upload is the default; mobile data upload is opt-in via Settings

5.5 Non-Functional Requirements

5.5.1 Performance

- App must launch and reach the Home screen within 3 seconds
- Heatmap must render within 4 seconds on a stable connection

5.5.2 Resource Constraints

- Background collection must consume less than 5% battery per hour

- Local storage is capped at 50MB with automatic deletion of synced records

5.5.3 Security

- The app must not collect name, email, IMEI, or IP address
- Backend API endpoints must require authentication tokens
- All data transmissions must be encrypted using TLS 1.2 or higher

5.5.4 Reliability

- No data loss during app crashes: all readings are written to local DB immediately
- Failed uploads are retried automatically every 30 minutes

5.5.5 Usability

- English and French language support based on device settings
- No more than 3 steps to enable or disable data collection

5.5.6 ML Accuracy

- The ML model must achieve at least 80% classification accuracy for coverage hole prediction
- Predictions require a minimum of 50 readings within 1 km squared to be valid